

MODULE 5 · WEEK 2

HuggingFace & Topic Modeling

Beyond counting words: embeddings that understand context, pre-trained transformers for summarization & zero-shot, and topics from text.

Lectures 57–63



This week

- 1 **From TF-IDF to transformers** — context, not just counts
- 2 **Embeddings** — turning text into meaningful vectors
- 3 **HuggingFace** — state-of-the-art models in a few lines
- 4 **Summarization & zero-shot** — pre-trained tasks out of the box
- 5 **Topic modeling** — discovering themes without labels

Why transformers beat TF-IDF

- TF-IDF counts words but doesn't understand language.
- "The bank was steep" vs. "the bank was closed" — same word, different meaning. Transformers get that.
- **HuggingFace** makes these models callable in a few lines — no deep-learning PhD required.

Key idea TF-IDF counts words; transformers understand them in context.

Embeddings

- An **embedding** maps a piece of text to a vector (e.g. 1×384) that captures meaning.
- Similar texts land close together; different ones land far apart.
- That vector becomes a feature you can cluster, search, or classify.

```
from sentence_transformers import SentenceTransformer
model = SentenceTransformer('all-MiniLM-L6-v2')
vecs = model.encode(texts)  # -> (n, 384)
```

Visualizing embeddings: 384 → 2 with PCA & UMAP

- You can't see 384 dimensions — reduce to 2 to plot.
- **PCA** finds the directions of most variance; **UMAP** preserves local neighborhoods.
- Plot the 2-D points and watch similar documents cluster together.

```
import umap
xy = umap.UMAP(n_components=2).fit_transform(vecs)
plt.scatter(xy[:,0], xy[:,1], c=labels)
```

Key idea Reducing embeddings to 2-D lets you literally see the structure in your text.

HuggingFace pipelines: summarization

- The pipeline API wraps a pre-trained model for a task in one call.
- Summarize long text, then **customize** length and style with parameters.
- No training required — you're standing on a model someone else trained.

```
from transformers import pipeline
summarizer = pipeline('summarization')
summarizer(text, max_length=60, min_length=20)
```

Zero-shot learning & sentiment

- **Zero-shot:** classify into labels the model was never trained on — you just name them.
- Sentiment analysis (positive/negative) comes out of the box.
- Great when you have no labeled data yet but need a prediction today.

```
clf = pipeline('zero-shot-classification')  
clf(text, candidate_labels=['billing', 'shipping', 'tech'])
```

Key idea Zero-shot lets you classify into brand-new categories with no training data.



Topic modeling

- **Unsupervised**: instead of predicting a label, discover the hidden themes in a document set.
- Combine embeddings with clustering to group documents by topic.
- Perfect for exploring a large text corpus before you decide what question to ask.



Lab this week

- Load a pre-trained HuggingFace model for embeddings/classification, compare it to your TF-IDF baseline, and try summarization and zero-shot.
- Reduce embeddings with PCA/UMAP to visualize and explore topics.



Key takeaways

- Transformers understand context; TF-IDF only counts words.
- Embeddings turn text into meaningful vectors you can cluster, search, and classify.
- PCA/UMAP reduce embeddings to 2-D so you can see the structure.
- HuggingFace pipelines give summarization, sentiment, and zero-shot in a few lines.
- Topic modeling discovers themes with no labels at all.